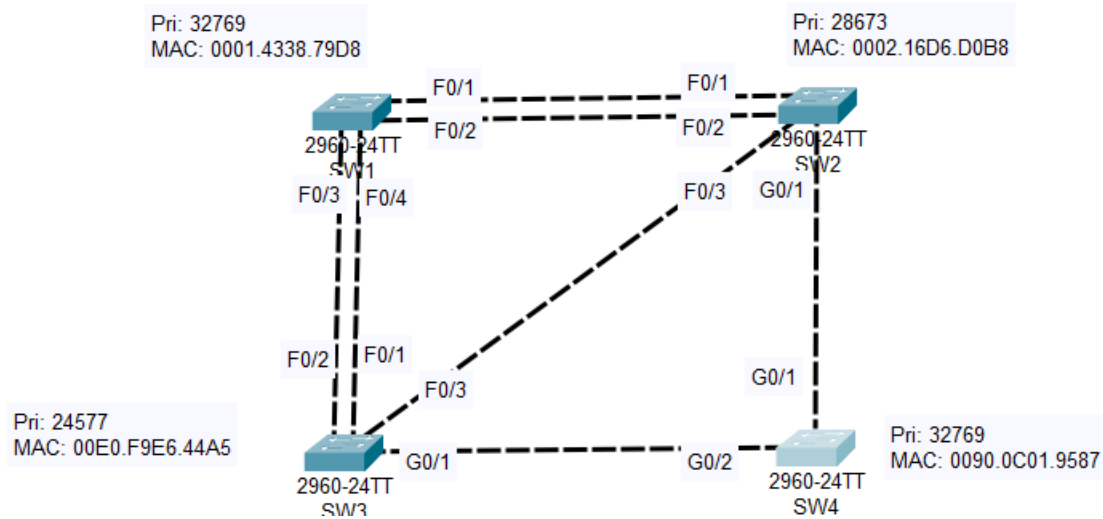




Topology

***TURN OFF LINK LIGHTS IN PACKET TRACER FOR THIS LAB - Options > Preferences > Show Link Lights**

1. Which switch is the root bridge?
- The root bridge is SW3.
2. Identify the role (root/designated/non-designated) of each switch port:

SW1	F0/1	Nondesignated
	F0/2	Nondesignated
	F0/3	Nondesignated
	F0/4	Root
SW2	F0/1	Designated
	F0/2	Designated
	F0/3	Nondesignated
	G0/1	Root
SW3	F0/1	Designated
	F0/2	Designated
	F0/3	Designated
	G0/1	Designated
SW4	G0/1	Designated
	G0/2	Root

Confirm your answers in the CLI

```
SW3#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     00E0.F9E6.44A5
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
             Address     00E0.F9E6.44A5
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Desg	FWD	19	128.3	P2p
Gi0/1	Desg	FWD	4	128.25	P2p

In this case, Bridge ID shows information about this switch, and Root ID shows information about this spanning tree's root bridge. Since both categories show the same information, SW3 IS the root bridge.

```
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     00E0.F9E6.44A5
             Cost         19
             Port         4(FastEthernet0/4)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0001.4338.79D8
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Altn	BLK	19	128.1	P2p
Fa0/2	Altn	BLK	19	128.2	P2p
Fa0/3	Altn	BLK	19	128.3	P2p
Fa0/4	Root	FWD	19	128.4	P2p

SW2#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 24577
 Address 00E0.F9E6.44A5
 Cost 8
 Port 25(GigabitEthernet0/1)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 28673 (priority 28672 sys-id-ext 1)
 Address 0002.16D6.D0B8
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/3	Altn	BLK	19	128.3	P2p
Fa0/1	Desg	FWD	19	128.1	P2p
Gi0/1	Root	FWD	4	128.25	P2p

SW4>en

SW4#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 24577
 Address 00E0.F9E6.44A5
 Cost 4
 Port 26(GigabitEthernet0/2)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 0090.0C01.9587
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/2	Root	FWD	4	128.26	P2p
Gi0/1	Desg	FWD	4	128.25	P2p